

Best Free Object Storage Options (No Debit Card)

Provider	Free Storage	Egress/Download Policy	API Access	Card Requirement
Azure for Students	\$100 credits + free Blob Storage quota	Free within credits	Full REST API, SDKs for Python/Java/.NET	✗ No card
Cloudflare R2	10 GB	Unlimited free egress	S3-compatible API	✗ No card
Backblaze B2	10 GB	Free egress up to 30 GB/month	S3-compatible API	✗ No card
Google Cloud Always Free	5 GB	100 GB free egress (NA only)	Cloud Storage API + Firebase SDK	✗ No card
Supabase Storage	1 GB	5 GB free egress/month	Integrated with Postgres + REST API	✗ No card

How One Can Use APIs

- **S3-Compatible APIs** Cloudflare R2, Backblaze B2, and IDrive e2 use the same API as AWS S3. One can write Python scripts with `boto3` or `s3fs` to upload/download files.

Task : Create a **bucket on Cloudflare R2** and write a Python app using `boto3` to store cricket match data.

Assignment: Cloud Object Storage & API Access

Part A – Supabase Setup

1. **Create a Supabase account** (free, no card required).
2. **Start a new project** → Supabase auto-provisions a Postgres database.
3. **Enable Storage** → Create a bucket named `cricket-data`.
4. **Upload sample files** (e.g., `match1.json`, `match2.json`) containing cricket match statistics.
5. **Generate API keys** → Copy the `SUPABASE_URL` and `SUPABASE_KEY`.

Part B – Supabase API Access

Write a **Python script** to upload and retrieve files:

```

import requests

SUPABASE_URL = "https://your-project.supabase.co"
SUPABASE_KEY = "your-service-key"

headers = {"apikey": SUPABASE_KEY, "Authorization": f"Bearer {SUPABASE_KEY}"}

# Upload file
with open("match1.json", "rb") as f:
    res = requests.post(
        f"{SUPABASE_URL}/storage/v1/object/cricket-data/match1.json",
        headers=headers,
        data=f
    )
print("Upload status:", res.status_code)

# Download file
res = requests.get(
    f"{SUPABASE_URL}/storage/v1/object/cricket-data/match1.json",
    headers=headers
)
print("File content:", res.text)

```

Part C – Boto3 Comparison

Then see how the same workflow looks in AWS S3 using **Boto3**:

```

import boto3

s3 = boto3.client('s3')
bucket = "cricket-data"

# Upload file
s3.upload_file("match1.json", bucket, "match1.json")

# Download file
s3.download_file(bucket, "match1.json", "local_match1.json")

```

Learning Outcomes

- Understand **object storage concepts** (bucket, object, API).
- Compare **Supabase REST API** vs **AWS Boto3 SDK**.
- Practice **secure API authentication** with keys/tokens.
- Build a **mini-project** : data storage + retrieval.